

Basic Civil & Mechanical Engineering
MID-II
Question Bank CSE-(A, B, C)

Short Answer Questions:-

Q1. List the various disciplines of civil engineering.

Answer: Civil engineering is divided into several main branches:

- Structural Engineering
- Geotechnical Engineering
- Transportation Engineering
- Water Resources Engineering
- Environmental Engineering

Q2. Classify the types of cement.

Answer: Cement is classified based on its properties and uses:

- Ordinary Portland Cement (OPC)
- Portland Pozzolana Cement (PPC)
- Rapid Hardening Cement
- Low Heat Cement
- Sulfate Resisting Cement

Q3. What is the primary role of a civil engineer?

Answer: The primary role of a civil engineer is to safely plan, design, construct, operate, and maintain public infrastructure—such as roads, bridges, buildings, water supply systems, and airports—while ensuring public safety and protecting the environment.

Q4. Name one method used for horizontal distance measurement in surveying.

Answer: Chaining (or Taping) is a standard direct method, which involves physically stretching a graduated chain or steel tape between two points to measure the distance.

Q5. Define Contour mapping.

Answer: Contour mapping is the process of drawing lines (called contour lines) on a map that connect points of equal height or elevation above a reference level (like mean sea level). It shows the shape, height, and steepness of the ground on a flat paper.

Q6. What is meant by traversing?

Answer: Traversing is a method of surveying where a series of connected straight lines (called traverse legs) are established on the ground. The lengths of these lines and the angles between them are measured to map out an entire area.

Q7. List out the different sources of water.

Answer: Water sources are divided into two main categories:

- **Surface Water:** Rivers, lakes, streams, ponds, and artificial storage reservoirs.
- **Groundwater:** Underground aquifers, open wells, tube wells, and natural springs.

Q8. What are the two main types of highway pavements?

Answer:

Flexible Pavement: Made of bituminous materials (asphalt/tar). These roads are flexible and transfer vehicle weight down through layers of soil.

Rigid Pavement: Made of Portland Cement Concrete (PCC) slabs. These roads are stiff, act like a solid block, and distribute weight over a very wide area.

Q9. Distinguish between a dam and a barrage.

Answer:

* **Dam:** A giant concrete or earthen wall built across a river primarily to store water in a huge upstream reservoir for long-term uses like generating electricity.

• **Barrage:** A structure built with a series of adjustable gates across a river primarily to divert water into irrigation canals by slightly raising the water level, without creating a huge storage lake.

Q10. What are the components of a railway track?

Answer: A standard railway track (also called the permanent way) consists of:

- Rails, Sleepers (Ties), Ballast, Subgrade

Q11. What are the characteristics of a good brick?

Answer: A good brick must have a uniform rectangular shape, a deep cherry-red color, high crushing strength, and a water absorption rate of less than 20%.

Q12. What is the primary objective of surveying?

Answer: The primary objective of surveying is to measure distances, angles, and elevations to prepare accurate maps or plans of an area.

Q13. List any four components of an airport.

Answer: The four main components are the runway, taxiway, apron, and terminal building.

Q14. What is hydrology and what is its importance?

Answer: Hydrology is the science that studies the occurrence and distribution of water on Earth; it is important for predicting floods and designing safe hydraulic structures like dams.

Long Answer Questions:-

Q1. Analyse the physical properties of cement.

Answer: The physical properties of cement determine its quality, strength, and durability in concrete structures. The key properties include:

- **Fineness:** It is a measure of the size of cement particles. Finer cement has a greater surface area, which increases the rate of hydration, leading to faster development of strength.
- **Consistency:** This refers to the ability of cement paste to flow. The *Standard Consistency test* (using Vicat's apparatus) finds the minimum water required to make a workable cement paste.
- **Setting Time:**
 - *** Initial setting time:** The time when cement begins to lose its plasticity (minimum 30 minutes).
 - **Final setting time:** The time when cement completely loses its plasticity and becomes hard (maximum 10 hours).
- **Soundness:** This refers to the ability of cement to retain its volume after hardening. Unsound cement undergoes excess expansion after setting due to lime or magnesia, causing cracking.
- **Compressive Strength:** It is the most critical structural property. It is tested by casting standard mortar cubes and testing them under a compressive testing machine after 3, 7, and 28 days of curing.

Q2. Explain in detail about the classification of aggregates.

Answer: Aggregates occupy 60% to 80% of a concrete mix and provide structural body. They are classified based on four main criteria:

1. Based on Size:

- *Fine Aggregates:* Particles that pass through a 4.75 mm IS sieve (e.g., river sand, manufactured sand).
- *Coarse Aggregates:* Particles retained on a 4.75 mm IS sieve up to 80 mm (e.g., crushed stone, gravel).

2. Based on Shape:

- *Rounded:* Have minimum voids (32-33%) and high workability, but low interlocking bond strength (e.g., river gravel).
- *Angular:* Have rough edges, high voids (38-45%), but create a strong mechanical interlocking bond, ideal for high-strength concrete.
- *Flaky and Elongated:* Thinner or longer than average; they reduce concrete strength and durability.

3. Based on Geological Origin:

- *Natural Aggregates:* Obtained from natural deposits by crushing rocks like granite, sandstone, or basalt.
- *Artificial Aggregates:* Manufactured artificially (e.g., blast furnace slag, expanded clay).

4. Based on Unit Weight:

- *Lightweight* (for insulation), *Normal weight* (standard structural concrete), and *Heavyweight* (for radiation shielding).

Q3. Explain the scope of transportation engineering and its importance.

Answer: Scope of Transportation Engineering: Transportation engineering deals with the application of scientific and technological principles to plan, functionally design, operate, and manage facilities for any mode of transportation. Its scope covers four major fields:

- **Highway Engineering:** Planning, geometric design, pavement design, and maintenance of roads.
- **Railway Engineering:** Planning and maintenance of track structures (permanent way), junctions, and signaling systems.
- **Airport Engineering:** Designing runways, taxiways, terminals, and air traffic control infrastructure.
- **Harbor and Dock Engineering:** Planning ports, channels, and lighthouses for water navigation.

Importance:

- **Economic Growth:** Facilitates the smooth transport of raw materials to industries and finished goods to markets.
- **Social Development:** Connects remote rural communities to cities, providing access to healthcare, education, and employment.
- **Defense and Emergency:** Crucial for rapid mobilization of defense forces during national emergencies or natural disasters.

Q4. Explain the importance of building construction and planning in civil engineering projects.

Answer: Building planning and construction management form the foundation of any successful civil engineering project. Their importance lies in the following areas:

- **Space Optimization:** Proper planning ensures effective room layout, circulation space, and maximum utilization of the available land footprint.
- **Structural Safety and Stability:** Careful construction planning ensures that the structure can safely handle live, dead, wind, and seismic loads by following standard building codes.
- **Cost and Time Efficiency:** Detailed scheduling (using tools like CPM/PERT) helps complete projects within the designated budget and avoids costly project delays.
- **Ventilation and Lighting (Prospect/Aspect):** Planning ensures the building receives natural sunlight and fresh air, reducing energy consumption for artificial lighting and cooling.
- **Aesthetics and Legal Compliance:** It ensures the building looks visually appealing while strictly obeying local municipal bye-laws, safety norms, and setback rules.

Q5. Explain the terms: i. True bearing, ii. Magnetic bearing iii. Whole circle bearing.

Answer: In surveying, a bearing is the horizontal angle measured between a reference line (meridian) and the survey line.

• i. True Bearing:

- It is the horizontal angle between the survey line and the True Meridian (the line connecting the geographic North and South poles).
- It remains constant over time and is determined through astronomical observations.

• ii. Magnetic Bearing:

- It is the horizontal angle between the survey line and the Magnetic Meridian (the direction indicated by a freely suspended magnetic needle).
- It changes continuously over time and location due to magnetic variations in the Earth.

• iii. Whole Circle Bearing (WCB):

- In this system, the bearing of a line is measured clockwise starting from the North meridian as the reference. Its value ranges anywhere from 0° to 360°. Prismatic compasses use the WCB system (e.g., Bearing of a line = 125°).

Q6. Define surveying. Explain the principles of surveying.

Answer: Definition: Surveying is the art and science of determining the relative positions of points above, on, or beneath the surface of the earth by means of direct or indirect measurements of distance, direction, and elevation.

Principles of Surveying:

1. To Work from Whole to Part:

○ Under this principle, a large area is first enclosed by a framework of highly accurate primary control stations. The area is then divided into smaller, less precise sections.

2. Location of a Point by Measurement from Two Control Points:

○ The relative position of a new station or point must always be fixed by taking at least two independent measurements (either linear or angular) from two already established reference points.

Q7. The following staff readings were observed successively with a level. 0.485, 1.210, 1.635, 3.395, 3. 775, 0.650, 1.400, 1. 795, 2.575, 3.375, 3.895, 1.735, 0.635, 1.605 m. The instrument has been moved after 5th and 11th readings. Determine the R.L. of various points, if the first reading was taken with a leveling staff held on a bench mark of R.L of 150 m using Height of Collimation method.

Level Book Table:

Station	BS (m)	IS (m)	FS (m)	HI (m)	RL (m)	Remarks
1	0.485			150.485	150.000	BM (Given)
2		1.210			149.275	
3		1.635			148.850	
4		3.395			147.090	
5	0.650		3.775	147.360	146.710	CP 1
6		1.400			145.960	
7		1.795			145.565	
8		2.575			144.785	
9		3.375			143.985	
10	1.735		3.895	145.200	143.465	CP 2
11		0.635			144.565	
12			1.605		143.595	Last Point
Total	$\Sigma BS = 2.870$		$\Sigma FS = 9.275$			

Arithmetic Check:

$$\Sigma BS - \Sigma FS = 2.870 - 9.275 = -6.405 \text{ m}$$

$$\text{Last RL} - \text{First RL} = 143.595 - 150.000 = -6.405 \text{ m}$$

Since $\Sigma BS - \Sigma FS = \text{Last RL} - \text{First RL}$, the calculations are correct.

Q8. List the importance of Transportation in a Nation's economic development.

Answer: Transportation is a critical infrastructure that facilitates the movement of people and goods, which is essential for economic development.

A good transportation system can help to:

- 1. Moves Goods and Services:** Transportation helps move products from factories to markets, supporting business and trade.
- 2. Creates Jobs:** It provides jobs for drivers, workers, and many others in related industries.
- 3. Boosts Industry:** Industries rely on transportation to get materials and deliver products.
- 4. Supports Farming:** Helps farmers transport crops to markets, ensuring food supply and income.
- 5. Improves Connectivity:** Makes it easier for people and businesses to connect across different regions.
- 6. Attracts Investment:** Good transportation systems attract businesses and investors.
- 7. Promotes Tourism:** Easier travel boosts tourism, creating economic benefits.
- 8. Increases Productivity:** Faster and cheaper transportation helps businesses work more efficiently.
- 9. Encourages Urban Growth:** Better transportation leads to the growth of cities and towns.
- 10. Strengthens Security:** Good transportation helps in emergencies by moving resources quickly.

Q9. What is meant by hydrology and state its importance.

Answer: Definition: Hydrology is the science that deals with the occurrence, circulation, distribution, and properties of water on the Earth's surface, underground, and in the atmosphere.

Terminology:-

- 1. Evaporation:** The process where liquid water from oceans, lakes, and rivers is heated by the sun and turns into water vapor (a gas) that rises into the air.
- 2. Transpiration:** The process where plants absorb water through their roots and release it as water vapor into the air through tiny holes in their leaves.
- 3. Condensation:** As the water vapor rises, it cools down and turns back into tiny liquid water droplets. These droplets bunch together to form clouds and fog.
- 4. Precipitation:** When the water droplets in the clouds become too heavy, they fall back to the Earth's surface as rain, snow, sleet, or hail.
- 5. Infiltration:** The process where rainwater hits the ground and soaks or seeps down into the soil to become groundwater.
- 6. Runoff:** The water that does not soak into the ground but instead flows over the land surface, emptying into streams, rivers, and eventually back into the ocean to start the cycle over again

Importance:-

Sustains Life: It continuously circulates the fresh water required for the survival of all plants, animals, and humans.

Powers Agriculture: It provides the natural rainfall necessary to grow crops and maintain global food supplies.

Regulates Climate: It moves heat around the planet through water vapor, keeping global temperatures balanced.

Fills Water Sources: It constantly recharges rivers, lakes, and underground aquifers used for drinking water.

Cleans the Environment: It naturally filters and purifies water as it moves through the evaporation and rain cycle

Q10. Outline the key considerations in the design and construction of tunnels.

Answer: Tunneling requires extensive engineering care due to unpredictable underground risks. The primary considerations include:

Geotechnical Investigation: Engineers must thoroughly test underground soil and rock conditions to determine the stability of the ground before digging.

Tunnel Alignment: The path must be carefully planned horizontally and vertically to avoid obstacles like deep foundations, utility lines, and heavy water bodies.

Excavation Method: The correct digging method—such as drilling and blasting for hard rock or using heavy Tunnel Boring Machines (TBM) for soft soil—must be chosen.

Structural Lining and Support: Temporary supports like shotcrete (sprayed concrete) and permanent concrete linings must be installed immediately to prevent the roof from collapsing.

Groundwater Control: Proper drainage systems and waterproofing membranes must be installed to prevent high-pressure underground water from leaking into the tunnel.

Ventilation and Lighting: Robust systems must be designed to constantly pump in fresh air, remove harmful vehicle exhaust fumes, and provide bright, continuous lighting for safety.

Environmental Impact: Construction must be managed carefully to minimize surface ground sinking, control noise, and safely dispose of massive amounts of excavated dirt.

Q11. Explain the basic components and functions of an Airport.

Answer: An airport infrastructure is divided into two distinct functional zones: **Airside** and **Landside**. The main components and their functions are:

- 1. Runways:** Long, flat paths where airplanes take off and land.
- 2. Taxiways:** Paths that connect runways with terminals, allowing aircraft to move around the airport.
- 3. Terminals:** Buildings where passengers check in, wait, and board their flights.
- 4. Control Tower:** A structure where air traffic controllers manage aircraft movements to ensure safe operations.
- 5. Gates:** Doorways in the terminal where passengers board and disembark from planes.
- 6. Parking Aprons:** Open areas where aircraft are parked, loaded, and unloaded.

7. **Cargo Facilities:** Areas for storing and handling goods transported by air.

8. **Security and Customs Areas:** Locations for security checks and passport control for international flights.

9. **Ground Services:** Support services like fueling, maintenance, baggage handling, and aircraft towing.

10. **Parking Areas:** Spaces for passengers to park their vehicles while they travel.